A defining feature of the TW-356 is its AI-based flight control and mission management system, which supports autonomous takeoff, landing, and route optimization even in adverse weather conditions. The aircraft's dualuse capability allows it to function as both a military ISR platform and a civilian logistics UAV, capable of transporting up to one ton of cargo to remote or disaster-stricken areas where traditional aircraft cannot operate.

In the defense domain, the TW-356 is deployed for border surveillance, maritime patrols, electronic signal monitoring, and communication relay operations, particularly in regions requiring persistent aerial presence. The UAV is equipped with sensor fusion technology that integrates visual, radar, and infrared data into a single operational interface, enhancing situational awareness for command centers.

Strategically, the Tengden TW-356 underscores China's growing leadership in heavy-class UAV development, blending endurance, payload capacity, and autonomy into one platform. It bridges the gap between traditional reconnaissance drones and transport aircraft, marking a step forward in the global trend toward large, multi-mission, AI-integrated UAV systems that redefine the operational potential of unmanned aviation.

Figure 5.8 presents the Tengden TW-356, a twin-boom, high-altitude long-endurance (HALE) UAV developed by Sichuan Tengden Technology Co., China. The figure highlights its twin-engine turboprop layout, highmounted wings, and wide-span airframe, which collectively contribute to its exceptional lift, endurance, and stability during extended missions. The central fuselage integrates multiple payload bays for ISR sensors, electronic warfare modules, or cargo pods, underscoring its dual military and civilian versatility. The nose section houses an EO/IR sensor turret, while underwing mounts support modular mission payloads. Designed with AI-based flight autonomy, the TW-356 can execute long-range surveillance